Data dictionary of `results/`

All result files are JSON (one file per benchmark problem unless stated otherwise). The field names and a few categorical labels were kept exactly as written by the analysis programs, which use Spanish names; this table gives their meaning. Fractions are given as numbers between 0 and 1 (e.g. `eajuste_med = 0.094` means 9.4 %) unless the name ends in `%`.

Common labels: `sistema` = problem; `segundos` = running time of that problem in seconds (on the machine that produced the file); `admisible` = admissible; `Stage 1` / `Stage 2` = stage of the selection; `Técnico` = technical failure (the nominal simulation fails); `salidas` = measured outputs; `estados` = state trajectories.

`selection/` and `settings/<variant>/` (analysis/selection.py)

Field	Meaning
`p`	number of estimated parameters of the PEtab problem
`S1`	subset returned by stage 1 (greedy selection by sensitivity energy)
`Rvar_S1_%`	R_var of the stage-1 subset (%)
`S1_cumple`	true if stage 1 reached R_var >= tau_R with at least 2 parameters
`cos_S1`, `erel_S1`	median cos(Delta) and e_rel of the stage-1 subset
`motivo_stage2`	why stage 2 was used ("stage 1 reaches R_var but is not admissible")
`etapa`	stage that produced the subset (`Stage 1`, `Stage 2`)
`S_antes_poda`	subset before backward elimination
`podados`	parameters removed by backward elimination
`S`	**final selected biological parameters**
`calibracion`	calibration parameters (scalings, offsets), always re-estimated with S
`Rvar_%`	R_var(S) (%)
`kappa`, `VIF`	condition number and largest VIF of the standardised columns of J for S
`cos_med`, `erel_med`	median over the scenarios of cos(Delta) and e_rel (no re-estimation)
`eajuste_med`	**median e_fit** (re-estimation of S and the calibration parameters)
`eajuste_cota`	upper bound e_rel used when e_fit was not computed (not used in the paper)
`eajuste_metodo`	present for the 8 problems where e_fit is computed to first order
`n_escenarios`	number of valid scenarios (of 15)
`admisible`	**admissible**:
`motivo`	reason when the problem could not be analysed
`T`, `etapa_paper`, `S_paper`	legacy fields of an earlier version of the analysis (state trajectories), not used

`external_validation/` (analysis/external_validation.py)

Field	Meaning
division	split: condiciones nuevas (held-out experimental conditions) or extrapolación temporal (t c) (later time points)
n_condiciones, n_train, n_test	number of conditions; training and test measurements
S, |S|, n_calibracion	subset validated, its size, number of calibration parameters
p_y	parameters that affect the outputs
locales_prueba, S_locales_prueba	parameters that only act in test conditions (excluded from the fits)
no_evaluable	why no training/test split was possible
A_chi2_test_red, A_chi2_test_full, A_chi2_test_theta0	real data: test chi2/n of the reduced model, of the full model (both re-fitted to the training data) and of the published theta0
A_chi2_train_red, A_chi2_train_full	real data: training chi2/n
B_chi2_test_theta0, B_chi2_test_red, B_chi2_test_full	synthetic data: test chi2/n (theta0 = signal-to-noise indicator)
B_epred_red, B_epred_full, B_epred_theta0	synthetic data: prediction error e_pred with respect to the known truth (1 for theta0)
B_error	failure of the synthetic experiment (simulation of the truth)
etapas_reanudadas	fits that were resumed from a checkpoint

robustness/ (analysis/robustness.py)

Field	Meaning
S, n_calibracion	subset (fixed) and number of calibration parameters
semilla, u	seed and the 15 random directions u(k) (15 x p), the same at every size
niveles	results per perturbation size ("1", "5", …, "50" = +-1 %, …, +-50 %)
niveles.*.n	valid scenarios
niveles.*.escenarios	indices k of the valid scenarios
niveles.*.eaj, erel, cos	e_fit, e_rel and cos(Delta) of every valid scenario
niveles.*.eaj_med, eaj_q25, eaj_q75, erel_med, cos_med	median and quartiles
patron	robustness pattern A-D (Results)
eajuste_metodo	linealizado (first order) or no lineal

single_parameter/ (analysis/single_parameter.py)

Field	Meaning
candidatos	number of candidates tested (up to 10 by energy and 10 by finite effect)
un_parametro	for each candidate: parametro, e_rel, e_fit, admisible
algun_parametro_solo_basta	true if some single parameter is admissible
mejor	best admissible single parameter j* (smallest e_fit)

Field	Meaning
`mejor_efit_10`, `S_efit_10`	median e_fit at +-10 % of {j*} and of the selected subset S
`mejor_validacion`	external validation of {j*} (same fields as external_validation)
`mejor_validacion_calibracion`	number of calibration parameters re-estimated with j* in that validation

Other files

File	Content
`simulator_check/simulator_check.jsonl`	one line per problem: `mediciones`, `condiciones`, `p`, `p_y`, `err_rel_mediana` and `err_rel_max_vs_simulations` (relative difference with the reference simulations of the benchmark), `nota` (why no comparison was possible)
`cost/simulation_times.json`	`p`, `t_sim` (seconds per simulation), `modo` (`salidas` or `estados`); Smith: `t_sim_medido` and `nota` (scaled from a second machine)
`crauste_blowup.json`	`nominal` and `un_parametro_1%` (24 one-at-a-time changes): `existe_en_[0,28]` (the solution exists on the measured horizon), `t_final` (time reached), `efectores_max`, `t_max`; `tolerancias`: number of failures (`fallos_*`) for each integrator tolerance
`sbml_to_lean/<problem>.json`	report of the translation to Lean: sizes (`n_estados`, `n_theta`, `reacciones`), checks done in Python before writing the Lean file, `lean` (file written), `teorema_final`, `traducido`
`sbml_to_lean/translation_fidelity.json`	for each model: `err_rel_max` (largest relative difference between the vector field of the Lean file and libRoadRunner), `err_theta`, `err_x0`, `err_por_punto`
`sct/sct_analysis.json`	SCT example: `theta0`, `R2_theta0`, `DE_100` (100 differential-evolution runs: seeds, R^2, parameter ranges, `cambia_signo` = changes sign), `sigma`, selection fields as above, `descartes` (rejected candidates and the parameter they are most collinear with), `niveles/patron` (robustness), `validacion` (chi2/n and R^2 on days 25-31; `red` = reduced, `full` = full, `_DE` = full model re-fitted by differential evolution; `estable_*` = stable)